Healthcare Analytics: Patient Readmission, Cost & Hospital Analysis

Final Data Analytics Project — SWYNEX Technologies

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Dataset Period: 2015–2024

1. Problem Statement

The project focuses on understanding key patterns in healthcare admissions from **2015–2024**.
The analysis examines patient characteristics, clinical conditions, readmissions, hospital capacity, and healthcare costs.
It aims to identify factors associated with **30-day readmissions** and variations in healthcare costs.
It also evaluates hospital bed capacity and patient admission patterns.
The findings provide insights for **better patient care, resource planning, and cost management**.

2. Key Analytical Questions

The analysis was guided by the following key questions:

1. What are the major patterns in patient admissions and hospital utilization?
2. Which factors are associated with higher 30-day readmission rates?
3. How do length of stay and healthcare costs vary across different categories?
4. What are the major patient and clinical patterns in the dataset?
5. How does hospital bed capacity relate to 30-day readmission rates?

3. Dataset Information

The data was sourced from **Kaggle** and covers healthcare records from **2015–2024**. It consists of five datasets: **Admissions, Billing, Patients, Hospitals, and Diagnoses**. These datasets were used to analyze patient admissions, readmissions, clinical factors, healthcare costs, and hospital capacity.

4. Data Cleaning & Preparation

The five datasets were cleaned and prepared using **Excel Power Query**. Data quality was checked using **Column Profiling** and other Power Query features to identify missing values, errors, duplicates, and inconsistencies. Data types and values were also reviewed and corrected to prepare the datasets for analysis.

5. Data Analysis

Exploratory Data Analysis (EDA) was performed using **Python, Pandas, and NumPy** to understand the datasets and identify important patterns and trends. The analysis included data exploration, statistical analysis, grouping and comparison of key variables, and visualization of important findings using charts.

6. Dashboard Development

An interactive healthcare analytics dashboard was developed using **Power BI** to present the key findings in a clear and meaningful way. The dashboard brings together patient, clinical, readmission, billing, cost, and hospital information through interactive visuals and filters, making it easier to explore trends, compare categories, and understand important healthcare patterns.

7. Key Business Insights

1. **Admission & Hospital Utilization:** Emergency admissions account for the largest share of total admissions, while hospital bed capacity varies widely across facilities.
2. **Readmission Factors:** Longer hospital stays, higher clinical severity, and ICU admissions are associated with higher 30-day readmission rates. The rate increases from 5.59% at Charlson Index 0 to 29.62% at Index 6.
3. **Cost & Length of Stay:** Higher-cost admissions generally have longer stays and higher readmission rates. The longest recorded stay was 90 days, with an associated cost of approximately ₹1.26 million.
4. **Patient & Clinical Patterns:** Cardiovascular is the most frequently recorded diagnosis category, while female and male patients are almost equally distributed.
5. **Bed Capacity & Readmission:** Hospital bed capacity shows little relationship with 30-day readmission rates, suggesting that bed capacity alone does not strongly explain readmission patterns.

8. Conclusion

The analysis provided an overall view of healthcare admissions, readmissions, patient characteristics, costs, and hospital capacity. The findings show that clinical severity, length of stay, and certain admission categories are important factors associated with readmission patterns, while hospital bed capacity alone has little relationship with readmissions. The project demonstrates how data analysis and visualization can support better understanding of healthcare performance and resource planning.

9. Tools Used

- **Excel** — Data cleaning and preparation
- **Python** — Exploratory data analysis and visualization
- **Power BI** — Data modeling, dashboard development, and interactive visualization

10. Dashboard

The final interactive healthcare analytics dashboard was developed in **Power BI** and consists of five pages covering executive overview, readmissions, patient and clinical analysis, billing and costs, and hospital analysis.

Dashboard Repository:
Dashboard